



CSIT-25-S1-02: X-Covid Insight

FYP-25-S1-08 User Requirement Specification (URS/SRS)

Supervisor

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1. Introduction

1.1 Overview

The COVID-19 pandemic has significantly impacted global well-being, making efficient and accurate diagnosis of the disease more critical than ever. With the rise of advanced technology, machine learning and computer vision have become a powerful tool in analysing medical imaging and detection of infections. Our product, X-COVID Insight, focuses on using those deep learning techniques to assist healthcare agencies in diagnosing COVID-19 through X-ray images.

The purpose of the User Requirement Specification document is to help assist the team in the approval of the X-COVID Insight project. This document will aid the team in the decision of the requirements of our project. Additionally, the features of each user will be implemented based on the requirements that we have decided on.

1.2 Intended Audience and Reading Suggestions

The User Requirement Specification document is intended for the people who will be involved in this project. This includes all the team members, along with both our supervisor, Mr Premrajan P. and our assessor, Mr Sionggo Japit. Mandatory knowledge of Information Technology terminology is required to read this document.

1.3 Project Scope

The main objective of this project is to build a role-based diagnostic platform that facilitates COVID-19 detection through Chest X-ray image classification. Our system is designed to simplify the operational levels at test centres, allowing COVID testers and test centres' admins to efficiently perform their respective responsibilities.

Our product provides patients with an online system to manage their covid-19 test appointments, reducing waiting time and the need for doctors. The system will have different tiers of subscriptions and will be made available for usage and viewing on computers.

2. Overall Description

2.1 Product Perspective

The product we are developing is a website application. This website application will be available for patients, doctors, and covid testers, test centre admins, system admins and unregistered users. The website allows the patients to book and cancel appointments. It also allows the patients to view their own x-ray records.

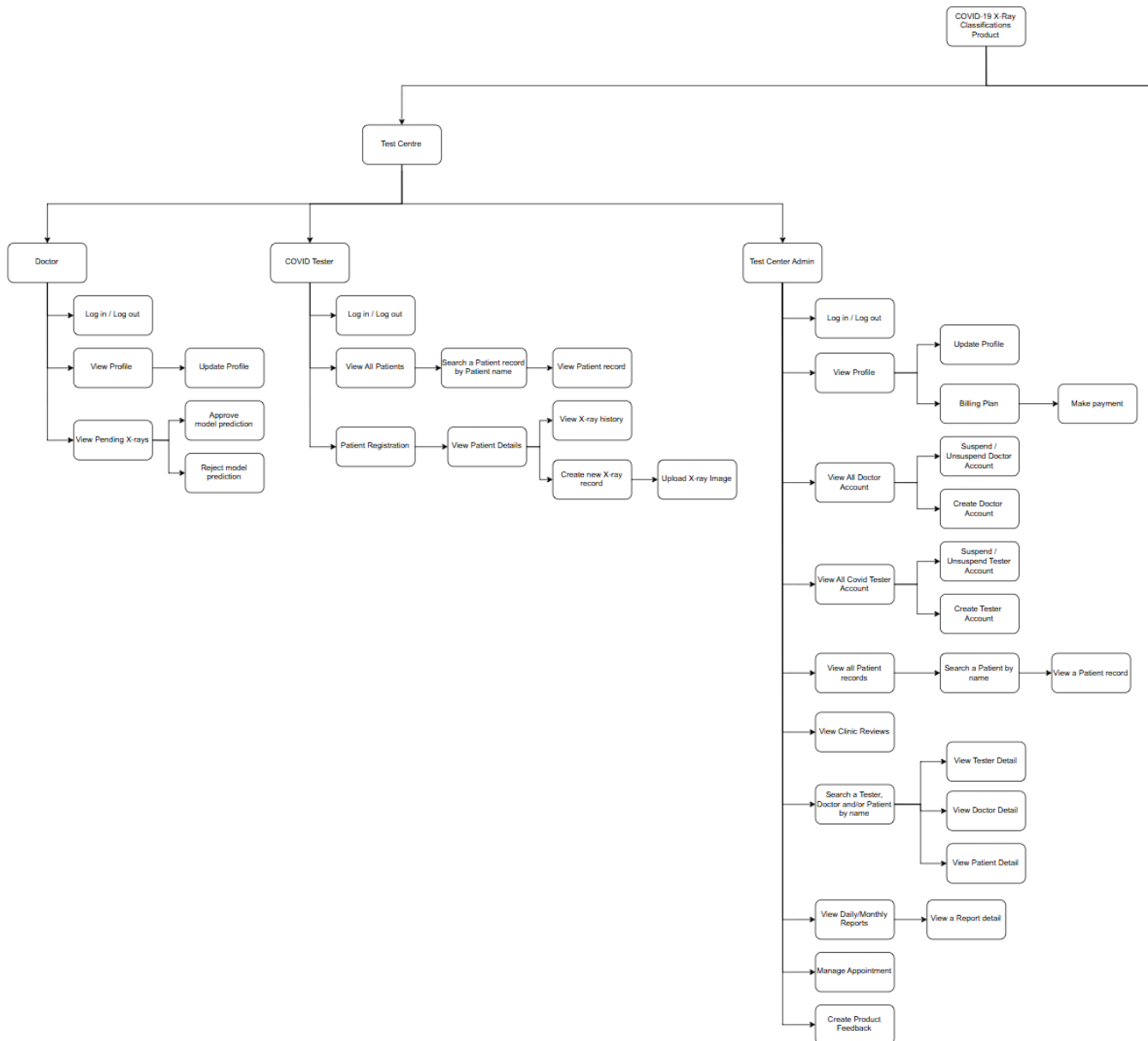
The testers are able to register patients as well as view the details for existing patients. They are also able to create a new x-ray record for patients and upload new x-ray images into their record. As for the doctors, they are able to view and validate the model's x-ray classification.

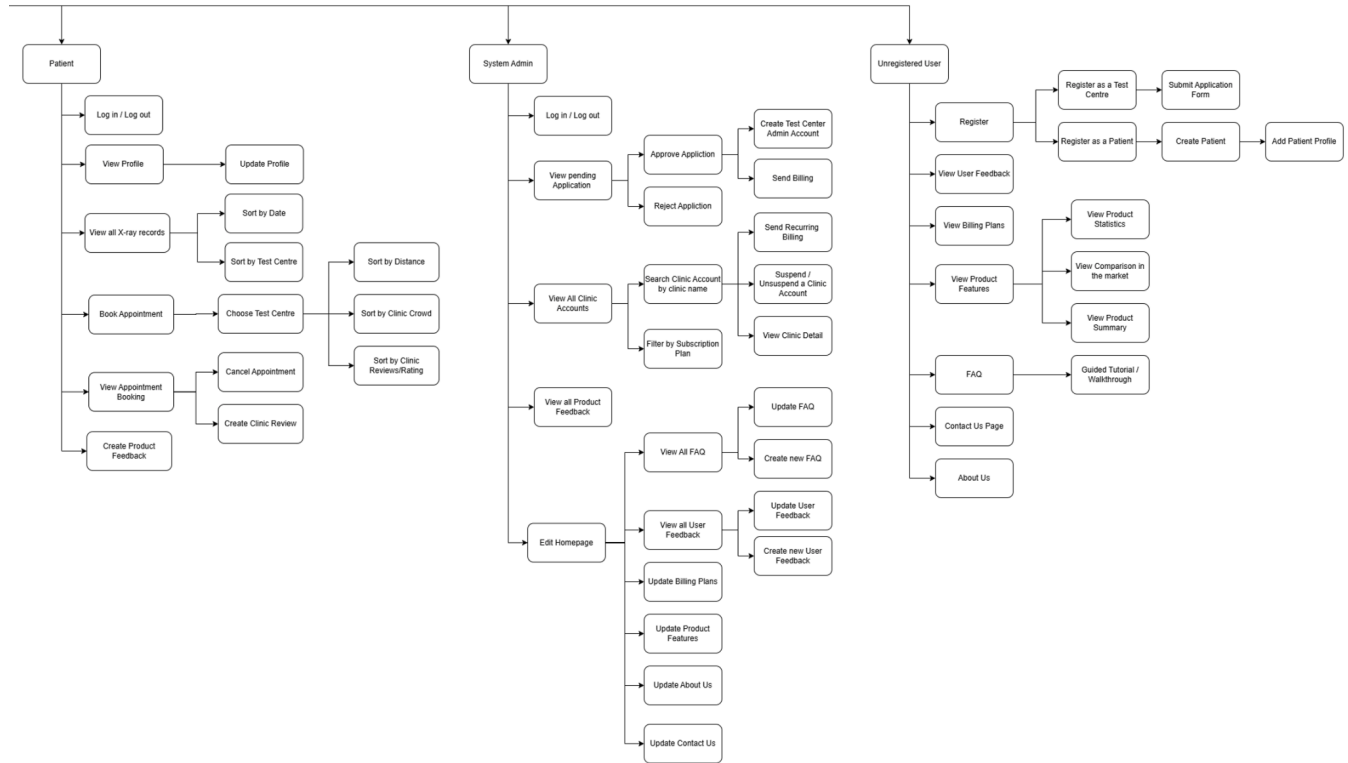
The test centre admin is able to create new doctor and tester accounts, suspend and unsuspend these accounts, and search and view the details of the accounts. Test centre admin also manages the payments and the product feedback.

The system admin approves and rejects test centre applications as well as creates test centre accounts. They are also able to suspend/ unsuspend clinic accounts. They can also send out the billing for test centres. Finally they are able to view product feedback and update the website.

The website includes features like product statistics and comparisons, product summary, an FAQ, a guided summary, a contact us page and an about us page. The website is also where users can register as a patient or a test centre.

2.2 Product Features





<u>Patient</u>	
FEATURE-1	Log In / Log Out
FEATURE-2	View Profile
FEATURE-3	Update Profile
FEATURE-4	View all X-ray Records
FEATURE-4.1	Sort Records by Date
FEATURE-4.2	Sort Records by Test Centres
FEATURE-5	Book Appointment
FEATURE-5.1	Choose Test Centre
FEATURE-5.1.1	Sort by Distance
FEATURE-5.1.2	Sort by Clinic Crowd
FEATURE-5.1.3	Sort by Clinic Rating
FEATURE-6	View Appointment Booking
FEATURE-6.1	Cancel Appointment
FEATURE-6.2	Create Clinic Review

<u>Doctor</u>	
FEATURE-1	Log In / Log Out
FEATURE-2	View Profile
FEATURE-3	Update Profile
FEATURE-4	View Pending X-rays
FEATURE-4.1	Approve Model Prediction
FEATURE-4.2	Reject Model Prediction

<u>Covid Tester</u>	
FEATURE-1	Log In / Log Out
FEATURE-2	View All Patients
FEATURE-3	Search by Patient Name
FEATURE-3.1	View Patient Record
FEATURE-4	Create Patient Registration
FEATURE-5	View Patient Profile Details
FEATURE-5.1	View X-ray History
FEATURE-5.2	Create a New X-ray Record
FEATURE-5.2.1	Upload X-ray Image

<u>Test Centre Admin</u>	
FEATURE-1	Log In / Log Out
FEATURE-2	View Profile
FEATURE-2.1	Update Profile
FEATURE-3	View and Manage Billing Plan
FEATURE-3.1	Make Payment
FEATURE-4	View Patient Record
FEATURE-4.1	Search Patient Name
FEATURE-4.1.2	View Patient Details
FEATURE-5	View Doctor Accounts
FEATURE-5.1	Suspend/Unsuspend Doctor Account
FEATURE-5.2	Create Doctor Account
FEATURE-6	View COVID Tester Account
FEATURE-6.1	Suspend/Unsuspend COVID Tester Account
FEATURE-6.2	Create COVID Tester Account
FEATURE-7	Search COVID Tester/Doctor/Patient by name
FEATURE-7.1	View COVID Tester/Doctor/Patient details
FEATURE-8	View All Reports
FEATURE-8.1	View Report Details
FEATURE-9	View Clinic Reviews
FEATURE-10	Create Product Feedback

<u>System Admin</u>	
FEATURE-1	Log In / Log Out
FEATURE-2	View Pending Application
FEATURE-2.1	Approve/Reject Application
FEATURE-3	Create Test Centre Admin Account
FEATURE-4	Send Billing/ Recurring Billing
FEATURE-5	View Clinic Accounts
FEATURE-5.1	Search Clinic Account
FEATURE-5.2	Filter Clinic by Subscription Plan
FEATURE-5.3	Suspend/Unsuspend Clinic Account
FEATURE-6	View Clinic Detail
FEATURE-7	View Product Feedback
FEATURE-8	View All User Feedback
FEATURE-9	Edit Homepage
FEATURE-9.1	View/Update/Create New FAQ
FEATURE-9.2	Update Billing Plans
FEATURE-9.3	Update Product Features
FEATURE-9.4	Update About Us
FEATURE-9.5	Update Contact Us

<u>Unregistered User</u>	
FEATURE-1	Register as Test Center
FEATURE-2	Submit Application
FEATURE-3	Register as Patient
FEATURE-3.1	Add Patient Profile
FEATURE-4	View User Feedback
FEATURE-5	View Billing Plans
FEATURE-6	View Product Statistics
FEATURE-7	View Product Comparison
FEATURE-8	View Product Summary
FEATURE-9	View FAQ
FEATURE-10	View Guided Walkthrough
FEATURE-11	View Contact Us Page
FEATURE-12	View About Us Page

2.3 User Classes and Characteristics

Users	Characteristics
Patient	<u>User Authentication & Profile Management:</u> • Can securely log in and out of the system. • Can view their profile details and update their profile information. <u>X-ray Records Management:</u> • Can view their past X-ray records • Can sort their records by Date and/or Test Centres. <u>X-ray Appointment Booking:</u> • Can book an X-ray appointment by choosing a preferred Test Centre based on: ○ Distance (find the nearest Centre to the Patient) ○ Clinic Crowd level (find the Centre with the highest availability) ○ Clinic Rating (ratings given by other patients) <u>X-ray Appointment Management:</u> • Can view their booked appointment. • Cancel the appointment if needed. • Can add a clinic rating after an X-ray.
Doctor	<u>User Authentication & Profile Management:</u> • Can securely log in and out of the system. • Can view their profile details and update their profile information. <u>X-ray Model Management:</u> • Can view pending X-ray images • Can approve or reject the system's X-ray model prediction.

Covid Tester	<u>User Authentication:</u> • Can securely log in and out of the system. <u>Patient Management & Patient Registration (Clinic side):</u> • Can view all records of booked patients. • Can search for a patient record using a patient name. • Can register a new patient upon arrival at the clinic. <u>X-ray Appointment Management:</u> • Can view patient profile details and the past X-ray records of a patient. • Can create a new X-ray record by uploading the new X-ray image to retrieve the X-ray model result.
Test Centre Admin	<u>User Authentication:</u> • Can securely log in and out of the system. • Can view their profile details and update their profile information. <u>Payment Management:</u> • Can view and manage billing plan • Can make payments <u>Patient Management</u> • Can view all records of booked patients. • Can search for a patient record using a patient name. • Can view specific patient details <u>Clinic Accounts Management:</u> • Can view all doctor and tester accounts • Can search for doctor and tester accounts • Can suspend/unsuspend all doctor and tester accounts • Can create doctor and tester accounts

	<u>Clinic Management:</u> • Can view all report details • Can view clinic reviews • Can create product feedback
System Admin	<u>User Authentication:</u> • Can securely log in and out of the system. <u>Application Management:</u> • Can view pending applications • Can approve/reject applications <u>Payment Management:</u> • Can send billing to test centres <u>Account Management</u> • Can create test centre accounts • Can view all clinic accounts • Can view clinic details • Can search for specific clinics • Can suspend/unsuspend clinic accounts • Can filter by subscription plan <u>Product Management:</u> • Can view product feedback <u>Homepage Management:</u> • Can edit homepage • Can view/update/create new FAQ • Can view user feedback • Can update billing plans, product features, about us and contact us page

Unregistered User	<u>Registration:</u> • Can register as test centre • Can register as patient <u>Viewing Further Details:</u> • Can view user feedback • Can view billing plans • Can view product in comparison with others in the market • Can view product statistics and summary • View FAQ and guided walkthrough • View about us and contact us page
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2.4 Operating Environment

Operating Environment 1	The website application requires a browser compatible with HTML, CSS, JavaScript, and PHP.
Operating Environment 2	The website application requires access to an internet connection.
Operating Environment 3	All personal data should be encrypted.

2.5 Design and Implementation Constraints

Constraint 1	UI should be created through HTML, CSS, JavaScript and PHP.
Constraint 2	Data should be stored in MongoDB.
Constraint 3	The website application requires access to a stable internet connection.
Constraint 4	For the best experience we recommend the Internet Explorer 11 or Mozilla Firefox 45 or Chrome 48.
Constraint 5	A secure database will be used to store the data of Patients, Doctors, Covid Testers, System Admin and Test Centre Admins.

2.6 User Documentation

Patient:

Upon entering the webpage, the user will be required to enter their login credentials. After logging in, the users will be able to:

- Book a New X-ray appointment based based on their filter or search criteria
- View past X-ray records for their own reference
- View upcoming appoints.
- Edit profile information to keep details up to date.
- Review their appointment experience to provide feedback to the test centers.

Doctor:

Upon entering the webpage, the user will be required to enter their login credentials. After logging in, users will be able to:

- View pending X-ray images for assessment.
- Approve or reject the model's prediction.
- Edit profile informatio to keep their own details up to date.

Covid Tester:

Upon entering the webpage, the user will be required to enter their login credentials. After logging in, users will be able to:

- View all upcoming patients for the day for better organisation across the centre.
- Register a patient for Covid testing.
- Search for a patient and access their profile & past records.
- Take and upload an X-ray image for analysis.
- Retrieve the model's prediction for diagnosis.

When the patient arrives at the test centre, the user will register the patient as present and then take the X-ray. Next, the user will upload the image and retrieve the model's predictions.

Test Center Admin:

Upon entering the webpage, the user will be required to enter their login credentials. After logging in, users will be able to:

- View their current billing plan and make bill payments. Additionally, they can also upgrade their billing plan as needed.
- Manage user accounts:
 - Viewing all Doctors and Covid Tester accounts.
 - Suspending and/or unsuspending accounts as required.
 - Creating new Covid Tester accounts.
- Accessing Patient records of those who have previously visited the test centre.

System Admin:

Upon entering the webpage, the user will be required to enter their login credentials. After logging in, users will be able to:

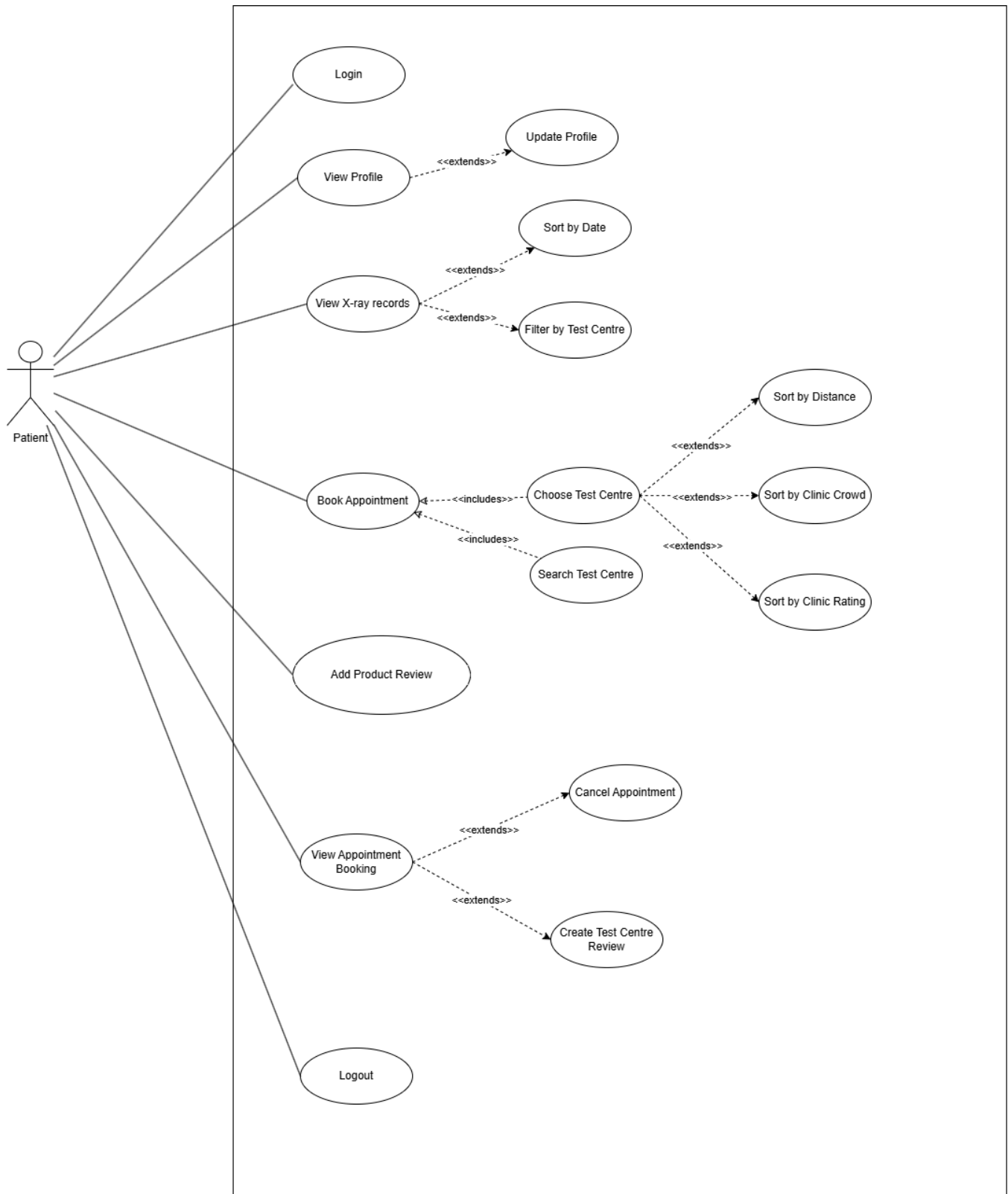
- Manage test centre applications by approving or rejecting new applications.
- Create new test centre admin accounts for approved centres
- Set up billing plans for different test centers
- Manage test centre (clinic) accounts
- Search test centre accounts by the search function.
- View product feedbacks to improve and enhance the system based on user feedbacks.
- Edit the product homepage to update or modify content such as:
 - Update billing plans page
 - Update FAQ page
 - Update product features page
 - Update about us page
 - Update contact us page

2.7 Assumption and Dependencies

Assumption 1	In order to have the best experience of using the web application we recommend our users to use either of the following Internet Explorer 11 or Mozilla Firefox 45 or Chrome 48.
Assumption 2	The web application demands for a stable internet connection.
Assumption 3	The web application also assumes that your Workstation is able to load the html files.
Assumption 4	The web application assumes that its users enter genuine information for proper functioning of the web application and prevent errors from arising.

3. System Features

3.1 Use Case Diagram - Patient



3.1.1 User Stories - Patient

#1	As a Patient, I want to log in the system so that I can access the system.
#2	As a Patient, I want to view my profile so that I can see my profile details.
#3	As a Patient, I want to update my profile so that I can update new profile information.
#4	As a Patient, I want to view all X-ray records so that I can see my X-ray results.
#5	As a Patient, I want to book an appointment so that I can select the clinic I want to visit.
#6	As a Patient, I want to view appointment booking so that I can cancel the appointment.
#7	As a Patient, I want to create a clinic review so that I can rate a clinic.
#8	As a Patient, I want to log out of the system so that I exit the system securely.
#9	As a Patient, I want to create product feedback so that I can give my suggestions.
#10	As a Patient, I want to search for a clinic so that I can choose any clinic I want.

3.1.2 Use Case Description - Patient

Name: Log In (Patient)		ID: P01
Stakeholders & Goals:	Patient can successfully log into the system.	
Description:	A Patient needs to log in before using the system.	
Actors:	Patient	
Pre-condition:	Patient has registered with the system.	
Trigger:	This event will be triggered when the user clicks on the Login button.	
Normal Flow:	1. Patient enters the login credentials & clicks the Login button. 2. System validates the login credentials. 3. Patient is redirected to the Patient site.	
Sub-Flow:	None	
Alternative/Exceptional Flow:	2a. System is unable to authenticate the user. 2b. System sends a prompt to the user to indicate that login is unsuccessful. 2c. Patient will have to re-enter login credentials & re-click the Login button.	

Name: View Profile (Patient)		ID: P02
Stakeholders & Goals:	Patient can successfully view their profile information.	
Description:	A Patient can view all their profile details.	
Actors:	Patient	
Pre-condition:	Patient is logged into the system.	
Trigger:	This event will be triggered when the user clicks on the View Profile button.	
Normal Flow:	1. System redirects Patient to the "View Profile Details" page.	
Sub-Flow 1:	2. Patient clicks on the "View X-ray result" button. 3. System directs Patient to the "X-ray result" page.	
Sub-Flow 2:	2. Patient clicks on the "Update Profile" button. 3. System directs Patient to the "Update Profile" page.	
Alternative/Exceptional Flow:	None	

Name: Update Profile (Patient)		ID: P03
Stakeholders & Goals:	Patient can successfully update their profile information.	
Description:	A Patient can retrieve and update new profile information on the system.	
Actors:	Patient	
Pre-condition:	Patient is logged into the system.	
Trigger:	This event will be triggered when the user clicks on the Update Profile button.	
Normal Flow:	1. System redirects Patient to the "Update Profile" page. 2. Patient updates their particulars and clicks the "Save new changes" button. 3. System validates the fields. 4. System prompts a success message "New information updated".	
Sub-Flow:	None	
Alternative/Exceptional Flow:	3a. System is unable to save new changes due to missing information. 3b. System prompts the user a error message "Please fill in all required information." 3c. Patient fills in the missing information and clicks the "Save new changes" button.	

Name: View X-ray records (Patient)		ID: P04
Stakeholders & Goals:	Patient can view all X-ray records in the system.	
Description:	A Patient can retrieve and see all past X-ray results.	
Actors:	Patient	
Pre-condition:	Patient has had at least 1 X-ray done before. Patient is logged into the system.	
Trigger:	This event will be triggered when the user clicks on the View X-ray Record button.	
Normal Flow:	1. System redirects Patient to the “View All X-ray Records” page.	
Sub-Flow 1: Sort records by Date	2. Patient clicks on the “Sort by Date” button. 3. System retrieves and sorts the records by Date (latest record first) in descending order.	
Sub-Flow 2: Filter records by Test Centre	2. Patient clicks on the “Sort by Test Centre” dropdown box, and selects the Clinic name. 3. System shows only records done by the Clinic selected.	
Alternative/Exceptional Flow:	None	

Name: Book Appointment		ID: P05
Stakeholders & Goals:	Patient can book an appointment to choose the test centre to go to.	
Description:	A Patient can choose the test centre to take the X-ray at.	
Actors:	Patient	
Pre-condition:	Patient is logged into the system.	
Trigger:	This event will be triggered when the user clicks on the Book Appointment button.	
Normal Flow:	1. System redirects Patient to the "Choose Test Centre" page. 2. System retrieves all available Test Centres. 3. Patient clicks on the Centre name.	
Sub-Flow Flow 1: Sort centre by Distance	3a. Patient clicks on the "Sort by Distance" button. 3b. System sorts the nearest 5 Test Centres based on the location. 3c. Patient clicks on the Centre name.	
Sub-Flow Flow 2: Sort centre by Availability	3a. Patient clicks on the "Sort by Clinic Crowd" button. 3b. System retrieves the 5 Test Centres with the lowest availability. 3c. Patient clicks on the Centre name.	
Sub-Flow Flow 3: Sort centre by Clinic Rating	3a. Patient clicks on the "Sort by Clinic Rating" button. 3b. System retrieves the 5 Test Centres with the highest ratings. 3c. Patient clicks on the Centre name.	
Alternative/Exceptional Flow:	None	

Name: View Appointment Booking (Cancel Appointment)		ID: P06
Stakeholders & Goals:	Patient can view booked appointments to cancel an appointment.	
Description:	A Patient can cancel the appointment booked.	
Actors:	Patient	
Pre-condition:	Patient has an upcoming appointment scheduled. Patient is on the View Appointment Booking page.	
Trigger:	This event will be triggered when the user clicks on the Cancel Appointment button.	
Normal Flow:	1. System prompts a confirmation message "Cancel the appointment?" 2. Patient clicks the "Yes" button. 3. System displays a success message "Appointment Cancelled. Please re-book a new appointment soon."	
Sub-Flow:	None	
Alternative/Exceptional Flow:	2a. Patient clicks the "No" button. 2b. Patient is redirected back to the View Appointment Booking page.	

Name: View Appointment Booking (Add Rating)		ID: P07
Stakeholders & Goals:	Patient can view booked appointments to add a clinic rating.	
Description:	A Patient can add a clinic rating and write a review.	
Actors:	Patient	
Pre-condition:	Patient has completed the appointment. Patient is on the View Appointment Booking page.	
Trigger:	This event will be triggered when the user clicks on the Add A Review button.	
Normal Flow:	1. System directs Patient to the "Add a Review" page. 2. Patient gives a rating out of 5 and clicks the "Submit review" button. 3. System displays a success message "Review Submitted Successfully. Thank you."	
Sub-Flow:	None	
Alternative/Exceptional Flow:	3a. System is unable to save new changes due to missing information. 3b. System prompts the user a error message "Please fill in all required information." 3c. Patient fills in the missing information and clicks the "Submit review" button.	

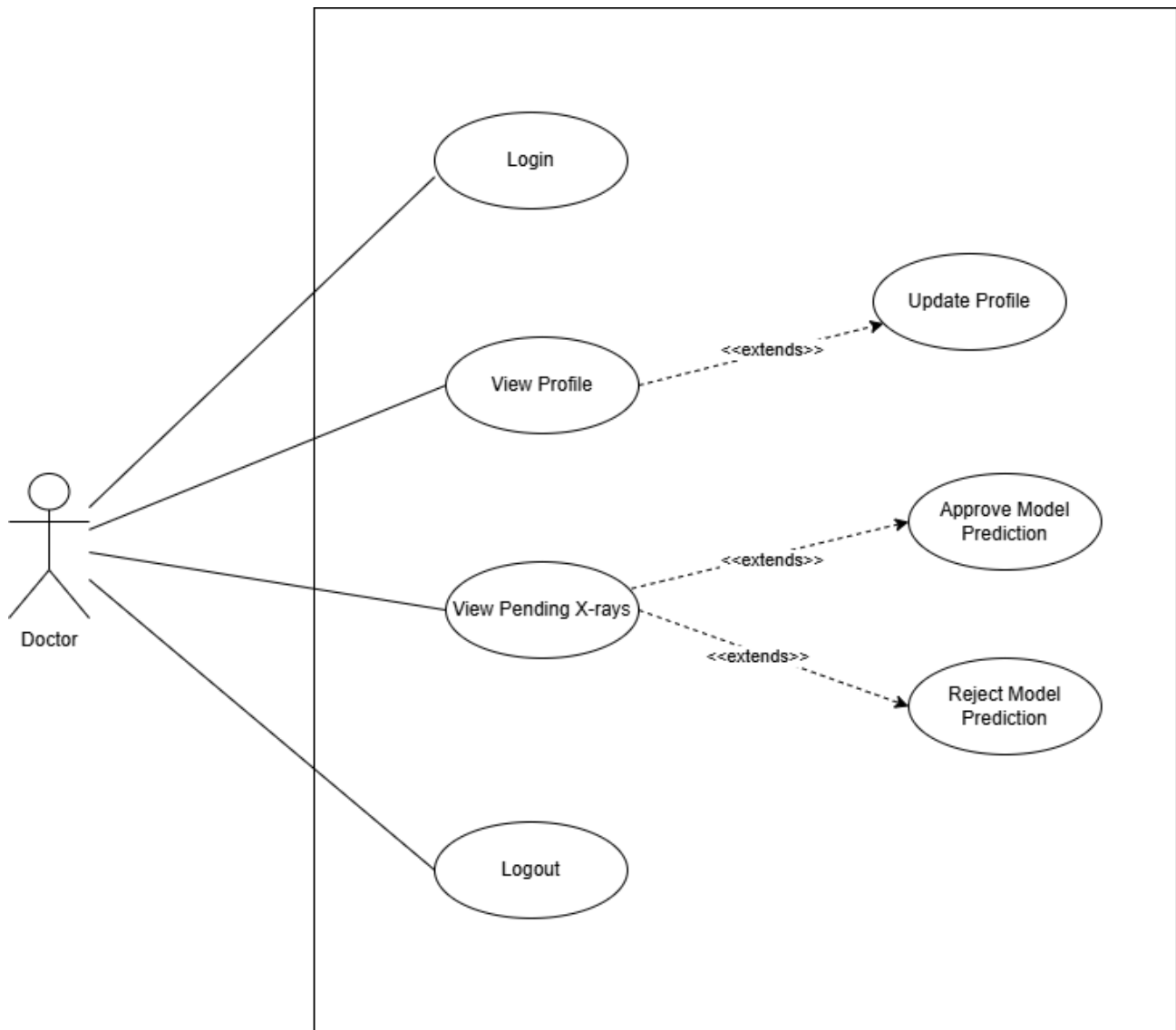
Name: Log Out (Patient)		ID: P08
Stakeholders & Goals:	Patient can successfully log out of the system.	
Description:	A Patient can log out of the system.	
Actors:	Patient	
Pre-condition:	Patient has logged into the system.	
Trigger:	This event will be triggered when the user clicks on the Logout button.	
Normal Flow:	1. System redirects the patient back to the "Login" page.	
Sub-Flow:	None	
Alternative/Exceptional Flow:	None	

Name: Add Product Feedback (Patient)		ID: P09
Stakeholders & Goals:	Patient can give personal feedback for the system.	
Description:	Patient need to add product feedback so that they can give suggestion for system to improve.	
Actors:	Patient	
Pre-condition:	Patient has logged into the system.	
Trigger:	Patient click Add product feedback option.	
Normal Flow:	1. Patient clicks on the "Add Product Feedback Button. 2. The system prompts the admin to enter rating and feedback details. 3. Patient enters the feedback and submits it.	

	4. The system saves the feedback and displays a confirmation a message.
Sub-Flow:	None
Alternative/Exceptional Flow:	3a.If the feedback submission fails, display an error message: "Unable to submit feedback. Please try again."

Name: Search a Clinic (Patient)		ID: T03
Stakeholders & Goals:	Patient can quickly locate a specific clinic by searching for a name.	
Description:	Patient needs to search target clinic so that can book appointment quickly.	
Actors:	Patient	
Pre-condition:	Patient has access to the clinic list.	
Trigger:	Patient click Search button.	
Normal Flow:	1. Patient types the clinic name in the search field and clicks the Search button. 2. The system searches the database for matching clinic 3. The system displays the matched clinic record(s).	
Sub-Flow:	None	
Alternative/Exceptional Flow:	2a. If no matching records are found, display the message: "No clinic records found."	

3.2 Use Case Diagram – Doctor



3.2.1 User Stories - Doctor

#1	As a Doctor, I want to log in the system so that I can access the system.
#2	As a Doctor, I want to view my profile so that I can see my profile details.
#3	As a Doctor, I want to update my profile so that I can update my new profile information.
#4	As a Doctor, I want to view pending X-rays so that I can validate the model's X-ray classification.
#5	As a Doctor, I want to log out of the system so that I exit the system securely.

3.2.2 Use Case Description - Doctor

Name: Log In (Doctor)		ID: P09
Stakeholders & Goals:	Doctor can successfully log into the system.	
Description:	A Doctor needs to log in before using the system.	
Actors:	Doctor	
Pre-condition:	Doctor has registered with the system.	
Trigger:	This event will be triggered when the user clicks on the Login button.	
Normal Flow:	1. Doctor enters the login credentials & clicks the Login button. 2. System validates the login credentials. 3. Doctor is redirected to the Doctor site.	
Sub-Flow:	None	
Alternative/Exceptional Flow:	2a. System is unable to authenticate the user. 2b. System sends a prompt to the user to indicate that login is unsuccessful. 2c. Doctor will have to re-enter login credentials & re-click the Login button.	

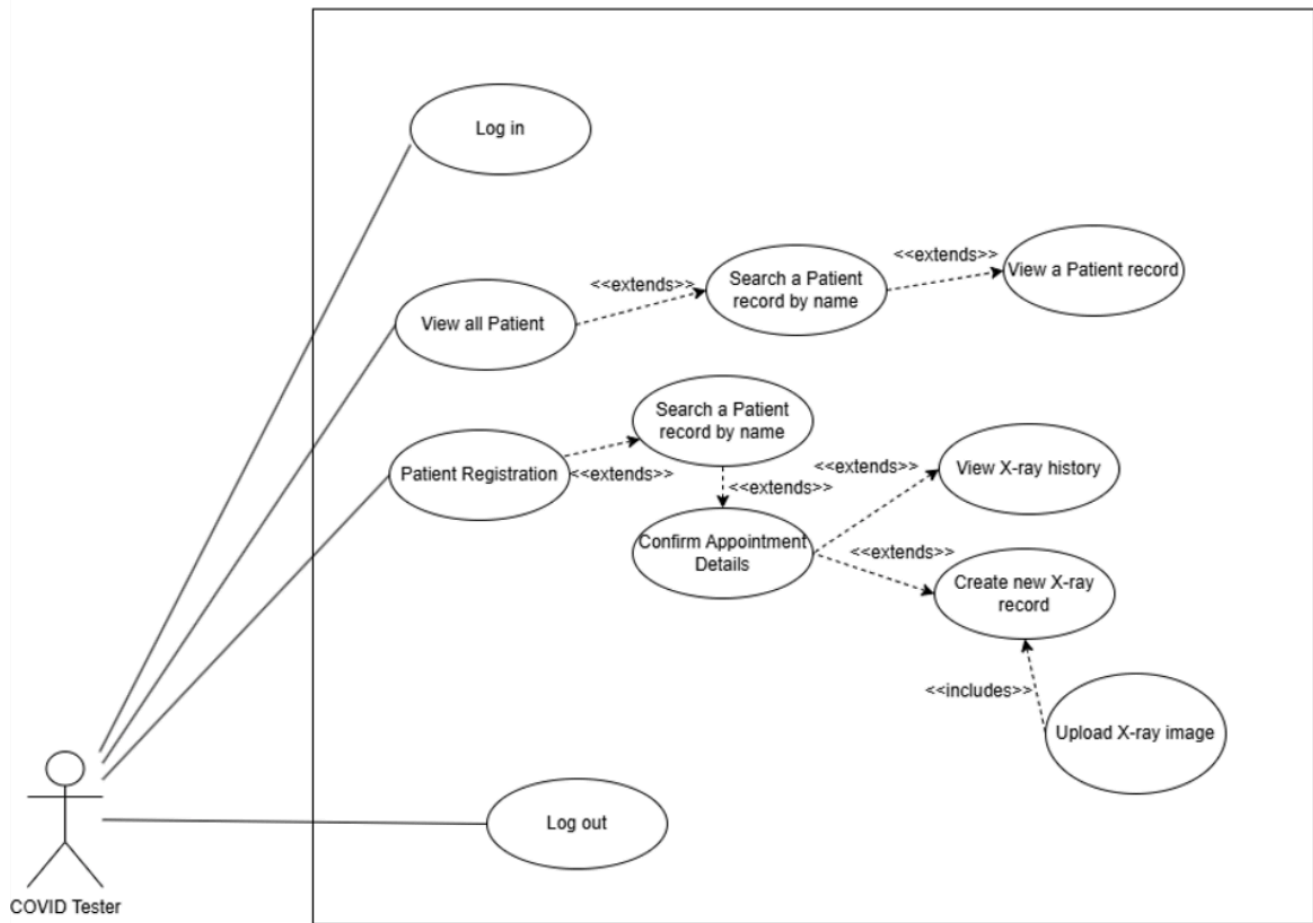
Name: View Profile (Doctor)		ID: P10
Stakeholders & Goals:	Doctor can successfully view their profile information.	
Description:	A Doctor can view all their profile details. Doctor is logged into the system.	
Actors:	Doctor	
Pre-condition:	None	
Trigger:	This event will be triggered when the user clicks on the View Profile button.	
Normal Flow:	1. System redirects Doctor to the “View Profile Details” page. 2. System retrieves and displays the Profile details.	
Sub-Flow:	None	
Alternative/Exceptional Flow:	None	

Name: Update Profile (Doctor)		ID: P11
Stakeholders & Goals:	Doctor can successfully update their profile information.	
Description:	A Doctor can retrieve and update new profile information on the system.	
Actors:	Doctor	
Pre-condition:	Doctor is logged into the system.	
Trigger:	This event will be triggered when the user clicks on the Update Profile button.	
Normal Flow:	1. System redirects Doctor to the "Update Profile" page. 2. Doctor updates their particulars and clicks the "Save new changes" button. 3. System validates the fields. 4. System prompts a success message "New information updated".	
Sub-Flow:	None	
Alternative/Exceptional Flow:	3a. System is unable to save new changes due to missing information. 3b. System prompts the user a error message "Please fill in all required information." 3c. Doctor fills in the missing information and clicks the "Save new changes" button.	

Name: View Pending X-rays		ID: P12
Stakeholders & Goals:	Doctor can view pending x-rays.	
Description:	A Doctor can retrieve and validate if the model correctly classifies a specific x-ray.	
Actors:	Doctor	
Pre-condition:	Doctor is logged into the system.	
Trigger:	This event will be triggered when the user clicks the View Pending X-rays button.	
Normal Flow:	1. System redirects Doctor to the "View Pending X-rays" page. 2. System retrieves a random X-ray image. 3. Doctor clicks on the "Approve Model Prediction" button if the classification is correct. 4. System prompts a success message "X-ray image validated".	
Sub-Flow:	3a. Doctor clicks on the "Disprove Model Prediction" button if the classification is incorrect. 3b. System redirects Doctor to the "Add reason for Disproving" page. 3c. Doctor fills in the reason and clicks the "Submit reason" button. 3d. Systems prompts a success message "Feedback received".	
Alternative/Exceptional Flow:	3ca. System validates the fields for missing information & sends an error prompt "Please fill in a reason". 3cb. Doctor fills in the required fields and clicks the "Submit reason" button.	

Name: Log Out (Doctor)		ID: P13
Stakeholders & Goals:	Doctor can successfully log out of the system.	
Description:	A Doctor can log out of the system.	
Actors:	Doctor	
Pre-condition:	Doctor is logged into the system.	
Trigger:	This event will be triggered when the user clicks on the Logout button.	
Normal Flow:	1. System redirects Doctor back to the “Login” page.	
Sub-Flow:	None	
Alternative/Exceptional Flow:	None	

3.3 Use Case Diagram - Covid Tester



3.3.1 User Stories - Covid Tester

#1	As a Covid Tester, I want to log in the system so that I can access the system.
#2	As a COVID Tester, I want to view all patients so that I can review the list.
#3	As a COVID Tester, I want to search for a patient by name so that I can quickly locate their information.
#4	As a COVID Tester, I want to view a patient record so that I can access and verify their detailed information.
#5	As a COVID Tester, I want to register a patient so that I can continue to diagnose.
#6	As a COVID Tester, I want to confirm appointment details so that I can understand their information.
#7	As a COVID Tester, I want to view a patient's X-ray history so that I can analyze past records for reference.
#8	As a COVID Tester, I want to create a new X-ray record for a patient so that I can document new findings.
#9	As a COVID Tester, I want to upload an X-ray image so that it can be added to the patient's record.
#10	As a COVID Tester, I want to log out of the system so that I can secure my account after use.
#11	As a COVID Tester, I want to search for appointments so that I can locate patients and upload new records.

3.3.2 Use Case Description - Covid Tester

Name: Log In (Covid Tester)		ID: T01
Stakeholders & Goals:	Covid Tester can successfully log into the system.	
Description:	A covid tester needs to log in so that can use the system.	
Actors:	Covid Tester	
Pre-condition:	Covid Tester has a valid account.	
Trigger:	Covid Tester clicks on the Login button.	
Normal Flow:	1. Covid Tester enter their information and click the login button. 1. System verifies the information with database. 2. If verify success, display Covid Tester homepage	
Sub-Flow:	None	
Alternative/Exceptional Flow:	2a. If the account is suspended, display a suspended message. 2b. If the password does not match, display invalid login message. 2c. If the account does not exist, display account does not exist message.	

Name: Log Out (Covid Tester)		ID: T10
Stakeholders & Goals:	Covid Tester can successfully log out the system.	
Description:	A covid tester needs to log out so that secure the system.	
Actors:	Covid Tester	
Pre-condition:	Covid Tester has logged in.	
Trigger:	Covid Tester clicks on the Log out button.	
Normal Flow:	1. Covid Tester clicks on the Log out button. 2. The system display the main home page.	
Sub-Flow:	None	
Alternative/Exceptional Flow:	None	

Name: View all Patients (Covid Tester)		ID: T02
Stakeholders & Goals:	Covid Tester can view all Patients.	
Description:	A covid tester wants to view all patients so that can retrieve all patient information.	
Actors:	COVID Tester	
Pre-condition:	COVID Tester has successfully logged into the system.	
Trigger:	Covid Tester clicks the "View All Patients" tab.	
Normal Flow:	1. COVID Tester clicks the "View All Patients" button. 2. The system retrieves the list of all patient records from the database. 3. The system displays the list of patients on the screen.	
Sub-Flow:	None	
Alternative/Exceptional Flow:	3a. If the system does not retrieve any records, display a no patient message. 3b. If the system cannot retrieve records, display an error message: "Unable to retrieve patient records."	

Name: Search a Patient Record by Name (Covid Tester)		ID: T03
Stakeholders & Goals:	COVID Tester can quickly locate a specific patient record by searching with the patient's name.	
Description:	A Covid tester needs to view the history Patient's X-ray image and retrieve the model's classification result.	
Actors:	COVID Tester	
Pre-condition:	COVID Tester has access to the patient records list.	
Trigger:	COVID Tester click Search button.	
Normal Flow:	5. COVID Tester types the patient name in the search field and clicks Search button. 6. The system searches the database for matching patient records 7. The system displays the matched patient record(s).	
Sub-Flow:	None	
Alternative/Exceptional Flow:	2a. If no matching records are found, display the message: "No patient records found."	

Name: View a Patient Record (Covid Tester)		ID: T04
Stakeholders & Goals:	The COVID Tester can view detailed information about a specific patient.	
Description:	A Covid tester needs to retrieve a Patient information.	
Actors:	COVID Tester	
Pre-condition:	COVID Tester has successfully located the patient record.	
Trigger:	COVID Tester clicks a specific patient record.	
Normal Flow:	1. COVID Tester clicks on a patient record. 2. The system retrieves the patient's details from the database. 3. The system displays the detailed patient record.	
Sub-Flow:	None	
Alternative/Exceptional Flow:	2a. If the system cannot retrieve the record, display an error message: "Unable to load patient details."	

Name: Patient Registration(Covid Tester)		ID: T05
Stakeholders & Goals:	COVID Tester can register the current patient.	
Description:	A COVID Tester needs to register the current patient to continue.	
Actors:	COVID Tester	
Pre-condition:	COVID Tester has successfully logged into the system.	
Trigger:	COVID Tester selects the "Register Patient" option from the menu.	
Normal Flow:	1. COVID Tester selects the "Present" option from the menu. 2. The system display registered successfully.	
Sub-Flow:	None	
Alternative/Exceptional Flow:	2a. If the system cannot registered, display an error message: "Unable to registered current patient"	

Name: Search an Appointment by Name (Covid Tester)		ID: T11
Stakeholders & Goals:	COVID Tester can quickly locate a specific appointment by searching with the patient's name.	
Description:	A Covid tester needs to upload the Patient's X-ray image and retrieve the model's classification result for the appointment.	
Actors:	COVID Tester	
Pre-condition:	COVID Tester has access to the patient records list.	
Trigger:	COVID Tester click Search button.	

Normal Flow:	1. COVID Tester types the patient name in the search field and clicks Search button. 2. The system searches the database for matching patient appointment. 3. The system displays the matched patient appointment(s).
Sub-Flow:	None
Alternative/Exceptional Flow:	2a. If no matching records are found, display the message: "No patient appointments found."

Name: Confirm Appointment Details(Covid Tester) ID: T06	
Stakeholders & Goals:	The COVID Tester can view detailed information.
Description:	A COVID Tester needs view detailed information.
Actors:	COVID Tester
Pre-condition:	COVID Tester has successfully registered the patient.
Trigger:	COVID Tester selects the "View Patient Details" option.

Normal Flow:	1. COVID Tester clicks on the "View Patient Details" button. 2. The system retrieves the patient's detailed information from the database. 3. The system displays the patient details on the screen.
Sub-Flow:	None
Alternative/Exceptional Flow:	2a. If the system cannot retrieve the details, display an error message: "Unable to load patient details."

Name: View X-ray History (Covid Tester)		ID: T07
Stakeholders & Goals:	COVID Tester can review the patient's X-ray history for reference.	
Description:	A COVID Tester needs to review the patient's X-ray history for reference.	
Actors:	COVID Tester	
Pre-condition:	COVID Tester has accessed the patient's details.	
Trigger:	COVID Tester selects the "View X-ray History" option.	

Normal Flow:	1. COVID Tester clicks on the "View X-ray History" button. 2. The system retrieves the X-ray history from the database. 3. The system displays the X-ray history.
Sub-Flow:	None
Alternative/Exceptional Flow:	2a. If no X-ray history is found, display the message: "No X-ray history available for this patient."

Name: Create New X-ray Record (Covid Tester)		ID: T08
Stakeholders & Goals:	COVID Tester can create a new X-ray record for the patient.	
Description:	A COVID Tester needs to create a new X-ray record for the patient.	
Actors:	COVID Tester	
Pre-condition:	COVID Tester has accessed the patient's details.	
Trigger:	COVID Tester selects the "Create New X-ray Record" option.	

Normal Flow:	1. COVID Tester clicks on the "Create New X-ray Record" button. 2. The system prompts the tester to enter necessary information for the X-ray record. 3. The system validates the information and saves the record. 4. The system displays a confirmation message.
Sub-Flow:	None
Alternative/Exceptional Flow:	2a. If required fields are incomplete, display an error message: "Please fill in all required fields."

Name: Upload X-ray Image (Covid Tester)		ID: T09
Stakeholders & Goals:	COVID Tester can upload X-ray images for image classification to get result.	
Description:	A COVID Tester needs to upload X-ray images for image classification to get result.	
Actors:	COVID Tester	
Pre-condition:	COVID Tester has created a new X-ray record.	
Trigger:	COVID Tester selects the "Upload X-ray Image" option.	

Normal Flow:	1. COVID Tester clicks on the "Upload X-ray Image" button. 2. The system prompts the tester to select an image file. 3. The tester selects the image file and confirms the upload. 4. The system displays the mprediction.
Sub-Flow:	None
Alternative/Exceptional Flow:	3a. If the file format is unsupported, display an error message: "Invalid file format. Please upload a valid X-ray image." 3b. If the upload fails, display an error message: "Image upload failed. Please try again."

3.4 Use Case Diagram - Test Centre Admin



3.4.1 User Stories – Test Centre Admin

#1	As a Test Center Admin, I want to log in the system so that I can access the system.
#2	As a Test Center Admin, I want to view my profile so that I can check and verify my personal and account details.
#3	As a Test Center Admin, I want to update my profile so that my information remains current.
#4	As a Test Center Admin, I want to view and manage my billing plan so that I can keep track of my subscriptions and payments.
#5	As a Test Center Admin, I want to make payments for my billing plan so that I can renew or maintain my subscription.
#6	As a Test Center Admin, I want to view all patient records so that I can manage patient data efficiently.
#7	As a Test Center Admin, I want to search for a patient by name so that I can quickly find their details.
#8	As a Test Center Admin, I want to view a specific patient's details so that I can review and manage their information.
#9	As a Test Center Admin, I want to view all doctor accounts so that I can manage their access and information.
#10	As a Test Center Admin, I want to suspend a doctor account so that I can restrict their access when necessary.
#11	As a Test Center Admin, I want to unsuspend a doctor account so that I can restore their access when needed.
#12	As a Test Center Admin, I want to create a doctor account so that new doctors can access the system.
#13	As a Test Center Admin, I want to view all COVID tester accounts so that I can manage their access and activity.
#14	As a Test Center Admin, I want to suspend a COVID tester account so that I can restrict their access when necessary.
#15	As a Test Center Admin, I want to unsuspend a COVID tester account so that I can restore their access when needed.
#16	As a Test Center Admin, I want to create a COVID tester account so that new testers can access the system.

#17	As a Test Center Admin, I want to search for testers, doctors, or patients by name so that I can quickly locate their information.
#18	As a Test Center Admin, I want to view a tester's details so that I can review and manage their information.
#19	As a Test Center Admin, I want to view a doctor's details so that I can manage their information and account.
#20	As a Test Center Admin, I want to view a patient's details so that I can manage their medical and personal information.
#21	As a Test Center Admin, I want to view all reports so that I can monitor operations and performance.
#22	As a Test Center Admin, I want to view a report detail so that I can track specific performance metrics.
#23	As a Test Center Admin, I want to view clinic reviews so that I can assess feedback and improve services.
#24	As a Test Center Admin, I want to create product feedback so that I can report issues or suggest improvements for the system.
#25	As a Test Center Admin, I want to log out of the system so that I can secure my account after use.
#26	As a Test Center Admin, I want to manage appointments so that I can efficiently schedule test bookings to ensure a well-arranged testing process.

3.4.2 Use Case Description – Test Centre Admin

Name: Log In (Test Center Admin)		ID: TA01
Stakeholders & Goals:	Test Center Admin can successfully log into the system.	
Description:	A Test Center Admin needs to successfully log into the system to use.	
Actors:	Test Center Admin	
Pre-condition:	Test Center Admin has a valid account.	
Trigger:	Test Center Admin clicks on the Login button.	
Normal Flow:	1. Test Center Admin enter their information and click the login button. 2. System verifies the information with database. 3. If verify success, display Test Center Admin homepage	
Sub-Flow:	None	
Alternative/Exceptional Flow:	2a. If the account is suspended, display a suspended message. 2b. If the password does not match, display invalid login message. 2c. If the account does not exist, display account does not exist message.	

Name: Log Out (Test Center Admin)		ID: TA25
Stakeholders & Goals:	Test Center Admin can successfully log out the system.	
Description:	A Test Center Admin needs to successfully log out the system.	
Actors:	Test Center Admin	
Pre-condition:	Test Center Admin has logged in.	
Trigger:	Test Center Admin clicks on the Log out button.	
Normal Flow:	1. Test Center Admin clicks on the Log out button. 2. The system display the main home page.	
Sub-Flow:	None	
Alternative/Exceptional Flow:	None	

Name: View Profile (Test Center Admin)		ID: TA02
Stakeholders & Goals:	Test Center Admin can view their profile to check personal and account details.	
Description:	Test Center Admin needs to view their profile to check personal and account details.	
Actors:	Test Center Admin	
Pre-condition:	Test Center Admin has successfully logged into the system.	
Trigger:	Test Center Admin selects the "View Profile" option.	
Normal Flow:	1. Test Center Admin clicks on the "View Profile" button. 2. The system retrieves the profile details from the database. 3. The system displays the profile information.	
Sub-Flow:	None	
Alternative/Exceptional Flow:	2a. If the system cannot retrieve the profile, display an error message: "Unable to load profile details."	

Name: Update Profile (Test Center Admin)		ID: TA03
Stakeholders & Goals:	Test Center Admin can update their profile information to keep it accurate.	
Description:	A Test Center Admin needs to update their profile information to keep it accurate.	
Actors:	Test Center Admin	
Pre-condition:	Test Center Admin has successfully accessed their profile.	
Trigger:	Test Center Admin selects the "Update Profile" option.	
Normal Flow:	1. Test Center Admin clicks on the "Update Profile" button. 2. The system displays editable profile fields. 3. Test Center Admin modifies the required fields. 4. The system validates the updates and saves the changes. 5. The system displays a confirmation message.	
Sub-Flow:	None	
Alternative/Exceptional Flow:	4a. If validation fails, display an error message: "Invalid input."	

Name: View Billing Plan (Test Center Admin)		ID: TA04
Stakeholders & Goals:	The Test Center Admin can view and manage their billing plan.	
Description:	A Test Center Admin needs to view and manage their billing plan.	
Actors:	Test Center Admin	
Pre-condition:	Test Center Admin has successfully accessed their profile.	
Trigger:	Test Center Admin selects the "Billing Plan" option.	
Normal Flow:	1. Test Center Admin clicks on the "Billing Plan" button. 2. The system retrieves the billing plan details. 3. The system displays the current plan and available options.	
Sub-Flow:	None	
Alternative/Exceptional Flow:	2a. If the system cannot retrieve the billing plan, display an error message: "Unable to load billing information."	

Name: Make Payment (Test Center Admin)		ID: TA05
Stakeholders & Goals:	Test Center Admin can make payments for their selected billing plan.	
Description:	A Test Center Admin needs to make payments for their selected billing plan.	
Actors:	Test Center Admin	
Pre-condition:	Test Center Admin has accessed the list of billing plans.	
Trigger:	Test Center Admin selects the "Make Payment" option of the target plan.	
Normal Flow:	1. Test Center Admin clicks on the "Make Payment" button of the target plan. 2. The system prompts the admin to enter payment details. 3. The admin submits the payment details. 4. The system processes the payment. 5. The system displays a confirmation message.	
Sub-Flow:	None	
Alternative/Exceptional Flow:	4a. If the payment fails, display an error message: "Payment unsuccessful. Please try again."	

Name: View All Doctor Accounts (Test Center Admin)		ID: TA09
Stakeholders & Goals:	Test Center Admin can view all doctor accounts to manage their information.	
Description:	A Test Center Admin needs to view all doctor accounts to manage their information.	
Actors:	Test Center Admin	
Pre-condition:	Test Center Admin has successfully logged into the system.	
Trigger:	Test Center Admin selects the "View All Doctor Accounts" option.	
Normal Flow:	1. Test Center Admin clicks on the "View All Doctor Accounts" button. 2. The system retrieves and displays the list of doctor accounts.	
Sub-Flow:	None	
Alternative/Exceptional Flow:	2a. If the system cannot retrieve the accounts, display an error message: "Unable to load doctor accounts." 2b. If no doctor account exist , display a message: "No doctor exist."	

Name: Suspend Doctor Account (Test Center Admin)		ID: TA10
Stakeholders & Goals:	Test Center Admin can suspend a doctor account when needed to manage.	
Description:	A Test Center Admin needs to suspend a doctor account when needed to manage.	
Actors:	Test Center Admin	
Pre-condition:	Test Center Admin has accessed the list of doctor accounts.	
Trigger:	Test Center Admin selects the "Suspend" option for a doctor account.	
Normal Flow:	1. Test Center Admin selects the "Suspend" option for a doctor account. 2. The system updates the account status in the database. 3. The system displays a confirmation message.	
Sub-Flow:	None	
Alternative/Exceptional Flow:	3a. If the update fails, display an error message: "Unable to update account status."	

Name: Unsuspend Doctor Account (Test Center Admin)		ID: TA11
Stakeholders & Goals:	Test Center Admin can unsuspend a doctor account when needed to manage.	
Description:	A Test Center Admin needs to unsuspend a doctor account when needed to manage.	
Actors:	Test Center Admin	
Pre-condition:	Test Center Admin has accessed the list of doctor accounts.	
Trigger:	Test Center Admin selects the "Unsuspend" option for a doctor account.	
Normal Flow:	4. Test Center Admin selects the "Unsuspend" option for a doctor account. 5. The system updates the account status in the database. 6. The system displays a confirmation message.	
Sub-Flow:	None	
Alternative/Exceptional Flow:	3a. If the update fails, display an error message: "Unable to update account status." 3b. If the account already be suspended, display error message: "Account already be suspended"	

Name: Create Doctor Account (Test Center Admin) ID: TA12	
Stakeholders & Goals:	Test Center Admin can create a new doctor account for system access.
Description:	A Test Center Admin needs to create a new doctor account for system access.
Actors:	Test Center Admin
Pre-condition:	Test Center Admin has accessed the doctor account list.
Trigger:	Test Center Admin selects the "Create Doctor Account" option.
Normal Flow:	1. Test Center Admin clicks on the "Create Doctor Account" button. 2. The system prompts the admin to enter the required details for the account. 3. The admin enters the details and submits the form. 4. The system validates the input and saves the new account. 5. The system displays a confirmation message.
Sub-Flow:	None
Alternative/Exceptional Flow:	1a. If the account limit has already reached, display a message: "Account limit reached." 4a. If validation fails, display an error message: "Please provide all required details." 4b. If the creation fails, display an error message: "Unable to create account."

Name: View All COVID Tester Accounts (Test Center Admin)		ID: TA13
Stakeholders & Goals:	Test Center Admin can view all COVID Tester accounts to check their information.	
Description:	A Test Center Admin needs to view all COVID Tester accounts to check their information.	
Actors:	Test Center Admin	
Pre-condition:	Test Center Admin has successfully logged into the system.	
Trigger:	Test Center Admin selects the "View All COVID Tester Accounts" option.	
Normal Flow:	1. Test Center Admin clicks on the "View All COVID Tester Accounts" button. 2. The system retrieves and displays the list of COVID Tester accounts.	
Sub-Flow:	None	
Alternative/Exceptional Flow:	2a. If the system cannot retrieve the accounts, display an error message: "Unable to load COVID Tester accounts." 2b. If no COVID Tester account exist , display a message: "No COVID Tester exist."	

Name: Suspend COVID Tester Account (Test Center Admin)		ID: TA14
Stakeholders & Goals:	Test Center Admin can suspend a COVID Tester account when needed to manage.	
Description:	A Test Center Admin needs to suspend a COVID Tester account when needed to manage.	
Actors:	Test Center Admin	
Pre-condition:	Test Center Admin has accessed the list of COVID Tester accounts.	
Trigger:	Test Center Admin selects the "Suspend" option for a COVID Tester account.	
Normal Flow:	1. Test Center Admin selects the "Suspend" option for a COVID Tester account. 2. The system updates the account status in the database. 3. The system displays a confirmation message.	
Sub-Flow:	None	
Alternative/Exceptional Flow:	3a. If the update fails, display an error message: "Unable to update account status."	

Name: Unsuspend COVID Tester Account (Test Center Admin)		ID: TA15
Stakeholders & Goals:	Test Center Admin can unsuspend a COVID Tester account when needed to manage.	
Description:	A Test Center Admin needs to unsuspend a COVID Tester account when needed to manage.	
Actors:	Test Center Admin	
Pre-condition:	Test Center Admin has accessed the list of COVID Tester accounts.	
Trigger:	Test Center Admin selects the "Unsuspend" option for a COVID Tester account.	
Normal Flow:	1. Test Center Admin selects the "Unsuspend" option for a COVID Tester account. 2. The system updates the account status in the database. 3. The system displays a confirmation message.	
Sub-Flow:	None	
Alternative/Exceptional Flow:	3a. If the update fails, display an error message: "Unable to update account status." 3b. If the account already be suspended, display error message: "Account already be suspended"	

Name: Create COVID Tester Account (Test Center Admin)		ID: TA16
Stakeholders & Goals:	Test Center Admin can create a new COVID Tester account for system access.	
Description:	A Test Center Admin needs to create a new COVID Tester account for system access.	
Actors:	Test Center Admin	
Pre-condition:	Test Center Admin has accessed the COVID Tester account list.	
Trigger:	Test Center Admin selects the "Create COVID Tester Account" option.	
Normal Flow:	1. Test Center Admin clicks on the "Create COVID Tester Account" button. 2. The system prompts the admin to enter the required details for the account. 3. The admin enters the details and submits the form. 4. The system validates the input and saves the new account. 5. The system displays a confirmation message.	
Sub-Flow:	None	
Alternative/Exceptional Flow:	1a. If the account limit has already reached, display a message: "Account limit reached." 4a. If validation fails, display an error message: "Please provide all required details." 4b. If the creation fails, display an error message: "Unable to create account."	

Name: View All Patient Records (Test Center Admin)		ID: TA06
Stakeholders & Goals:	Test Center Admin can view all patient records to check patient details.	
Description:	A Test Center Admin needs to view all patient records to check patient details.	
Actors:	Test Center Admin	
Pre-condition:	Test Center Admin has successfully logged into the system.	
Trigger:	Test Center Admin selects the "View All Patient Records" option.	
Normal Flow:	1. Test Center Admin clicks on the "View All Patient Records" button. 2. The system retrieves all patient records from the database. 3. The system displays the list of all patient records.	
Sub-Flow:	None	
Alternative/Exceptional Flow:	2a. If no patient record exists, display a message: "No patient record exists." 2a. If the system cannot retrieve the records, display an error message: "Unable to load patient records."	

Name: Search a Patient (Test Center Admin)		ID: TA07
Stakeholders & Goals:	Test Center Admin can search for a patient by name to quickly locate their record.	
Description:	A Test Center Admin needs to search for a patient by name to quickly locate their record.	
Actors:	Test Center Admin	
Pre-condition:	Test Center Admin has accessed the patient records list.	
Trigger:	Test Center Admin enters a patient's name in the search bar and click the search option.	
Normal Flow:	1. Test Center Admin enters the patient's name in the search bar. 2. The system searches the database for matching records. 3. The system displays the matching patient record(s).	
Sub-Flow:	None	
Alternative/Exceptional Flow:	2a. If no matching records are found, display the message: "No patient records found."	

Name: View a Patient Record (Test Center Admin)		ID: TA08
Stakeholders & Goals:	Test Center Admin can view detailed information about a specific patient.	
Description:	A Test Center Admin needs to view detailed information about a specific patient.	
Actors:	Test Center Admin	
Pre-condition:	Test Center Admin has located the specific patient record.	
Trigger:	Test Center Admin clicks the patient record from the list or search results.	
Normal Flow:	1. Test Center Admin clicks on a patient record. 2. The system retrieves the patient's details from the database. 3. The system displays the detailed patient record.	
Sub-Flow:	None	
Alternative/Exceptional Flow:	2a. If the system cannot retrieve the record, display an error message: "Unable to load patient details."	

Name: Search a Tester/Doctor/ Patient (Test Center Admin)		ID: TA17
Stakeholders & Goals:	Test Center Admin can search for testers, doctors, or patients by name to locate their information.	
Description:	A Test Center Admin needs to search for testers, doctors, or patients by name to locate their information.	
Actors:	Test Center Admin	
Pre-condition:	Test Center Admin has successfully logged into the system.	
Trigger:	Test Center Admin clicks on the Search button.	
Normal Flow:	1. Test Center Admin enters the name in the search bar.	
Sub-Flow Flow 1: Search Tester	2. Test Center Admin clicks the dropdown box and clicks Tester. 3. The system searches the database for matching records. 4. The system displays the matching Tester results.	
Sub-Flow Flow 2: Search Doctor	2. Test Center Admin clicks the dropdown box and clicks Doctor. 3. The system searches the database for matching records. 4. The system displays the matching Doctor results.	
Sub-Flow Flow 1: Search Patient	2. Test Center Admin clicks the dropdown box and clicks Patient. 3. The system searches the database for matching records. 4. The system displays the matching Patient results.	
Alternative/Exceptional Flow:	2a. If no records are found, display the message: "No matching records found." 2b. If the system cannot retrieve reviews , display an error message: "Unable to search ."	

Name: View a Tester Detail (Test Center Admin)		ID: TA18
Stakeholders & Goals:	Test Center Admin can view the detailed information of a tester to manage.	
Description:	A Test Center Admin needs to view the detailed information of a tester to manage.	
Actors:	Test Center Admin	
Pre-condition:	Test Center Admin has located the tester's record.	
Trigger:	Test Center Admin selects a tester record from the search results.	
Normal Flow:	1. Test Center Admin clicks on the tester record. 2. The system retrieves the tester's details from the database. 3. The system displays the detailed information.	
Sub-Flow:	None	
Alternative/Exceptional Flow:	2a. If the system cannot retrieve the record, display an error message: "Unable to load tester details."	

Name: View a Doctor Detail (Test Center Admin)		ID: TA19
Stakeholders & Goals:	Test Center Admin can view the detailed information of a Doctor to manage.	
Description:	A Test Center Admin needs to view the detailed information of a Doctor to manage.	
Actors:	Test Center Admin	
Pre-condition:	Test Center Admin has located the Doctor's record.	
Trigger:	Test Center Admin selects a Doctor record from the search results.	
Normal Flow:	1. Test Center Admin clicks on the Doctor record. 2. The system retrieves the Doctor's details from the database. 3. The system displays the detailed information.	
Sub-Flow:	None	
Alternative/Exceptional Flow:	2a. If the system cannot retrieve the record, display an error message: "Unable to load Doctor details."	

Name: View a Patient Detail (Test Center Admin)		ID: TA20
Stakeholders & Goals:	Test Center Admin can view the detailed information of a Patient to manage.	
Description:	A Test Center Admin needs to view the detailed information of a Patient to manage.	
Actors:	Test Center Admin	
Pre-condition:	Test Center Admin has located the Patient's record.	
Trigger:	Test Center Admin selects a Patient record from the search results.	
Normal Flow:	1. Test Center Admin clicks on the Patient record. 2. The system retrieves the Patient's details from the database. 3. The system displays the detailed information.	
Sub-Flow:	None	
Alternative/Exceptional Flow:	2a. If the system cannot retrieve the record, display an error message: "Unable to load Patient details."	

Name: View Reviews (Test Center Admin)		ID: TA23
Stakeholders & Goals:	Test Center Admin can view clinic reviews to assess feedback and improve services.	
Description:	A Test Center Admin needs to access and review feedback left by customers to identify areas of improvement.	
Actors:	Test Center Admin	
Pre-condition:	Test Center Admin has successfully logged into the system.	
Trigger:	Test Center Admin selects the "View Reviews" option.	
Normal Flow:	1. Test Center Admin clicks on the "View Reviews" button. 2. The system retrieves a list of clinic reviews from the database. 3. The system displays these clinic reviews.	
Sub-Flow:	None	
Alternative/Exceptional Flow:	2a. If no reviews are available, display a message: "No reviews available." 2b. If there is a system error in fetching reviews, display an error message: "Unable to retrieve reviews."	

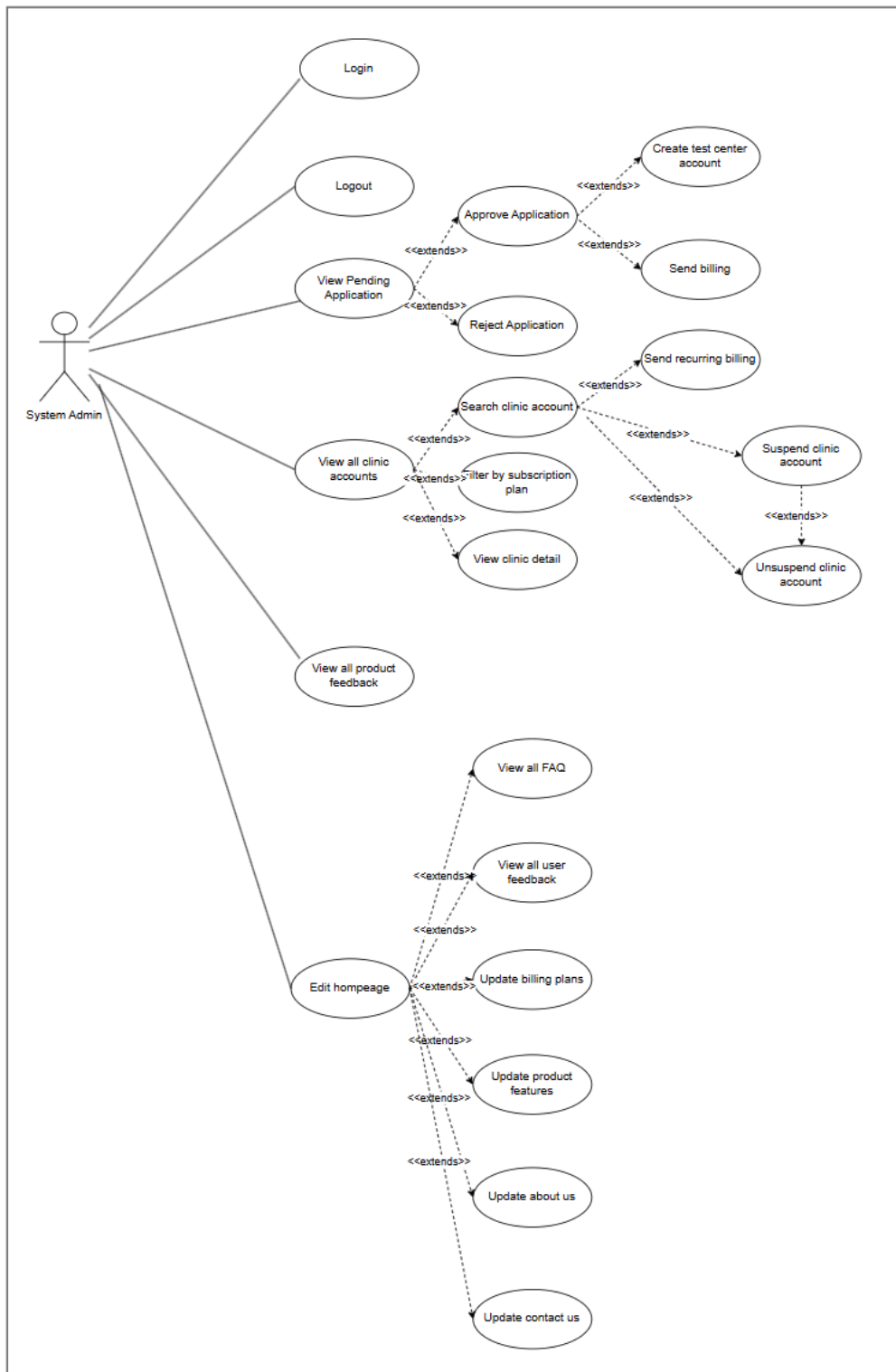
Name: View daily/monthly Reports (Test Center Admin)		ID: TA21
Stakeholders & Goals:	The Test Center Admin can view all available reports to monitor and analyze operational data.	
Description:	A Test Center Admin needs to view all available reports to monitor and analyze operational data.	
Actors:	Test Center Admin	
Pre-condition:	Test Center Admin has successfully logged into the system.	
Trigger:	Test Center Admin selects period and clicks on the "View Reports" button.	
Normal Flow:	1. Test Center Admin selects period and clicks on the "View Reports" button	
Sub-Flow Flow 1: Daily report	2. The system retrieves a list of available daily reports. 3. The system displays the list of daily reports	
Sub-Flow Flow 2: Monthly report	2. The system retrieves a list of available monthly reports. 3. The system displays the list of monthly reports	
Alternative/Exceptional Flow:	2a. If no reports are available, display the message: "No reports available." 2b. If the system cannot retrieve the report, display an error message: "Unable to load report."	

Name: View a Report Detail (Test Center Admin)		ID: TA22
Stakeholders & Goals:	Test Center Admin can view a report detail to track specific performance metrics.	
Description:	Test Center Admin needs to view a report detail to track specific performance metrics.	
Actors:	Test Center Admin	
Pre-condition:	Test Center Admin has accessed the list of reports.	
Trigger:	Test Center Admin selects the specific report.	
Normal Flow:	1. Test Center Admin selects the specific report. 2. The system retrieves the report data. 3. The system displays the report.	
Sub-Flow:	None	
Alternative/Exceptional Flow:	2a. If the system cannot retrieve the report, display an error message: "Unable to load report."	

Name: Create Product Feedback (Test Center Admin)		ID: TA24
Stakeholders & Goals:	Test Center Admin can create product feedback to report issues or suggest improvements.	
Description:	A Test Center Admin needs to create product feedback to report issues or suggest improvements.	
Actors:	Test Center Admin	
Pre-condition:	Test Center Admin has successfully logged into the system.	
Trigger:	Test Center Admin selects the "Create Product Feedback" option.	
Normal Flow:	1. Test Center Admin clicks on the "Create Product Feedback" button. 2. The system prompts the admin to enter rating and feedback details. 3. Test Center Admin enters the feedback and submits it. 4. The system saves the feedback and displays a confirmation message.	
Sub-Flow:	None	
Alternative/Exceptional Flow:	3a. If the feedback submission fails, display an error message: "Unable to submit feedback. Please try again."	

Name: Manage Appointments (Test Center Admin) ID: TA26	
Stakeholders & Goals:	Test Center Admin can manage appointments efficiently to ensure a well-arranged testing process.
Description:	A Test Center Admin needs to schedule appointments to manage test center operations effectively.
Actors:	Test Center Admin
Pre-condition:	Test Center Admin has successfully logged into the system.
Trigger:	Test Center Admin selects the "Manage Appointment" option.
Normal Flow:	1. Test Center Admin clicks on the "Manage Appointment" button. 2. The system displays the time slot management of the current appointment list. 3. Test Center Admin modify the time slot as needed. 4. The system validates and stores the time slot information in the database. 5. The system displays a confirmation page.
Sub-Flow:	None
Alternative/Exceptional Flow:	4a. If scheduling fails due to time conflicts, display an error message: "Modified time slot is unavailable." 4b. If the appointment management submission fails, display an error message: "Unable to manage appointment. Please try again."

3.5 Use Case Diagram – System Admin



3.5.1 User Stories - System Admin

#1	As a System Admin, I want to log in the system so that I can access the system.
#2	As a System Admin, I want to view pending application so that I can see all the applied test center
#3	As a System Admin, I want to approve application if they are valid test center
#4	As a System Admin, I want to reject application if the test center is not valid
#5	As a System Admin , I want to create test center admin account so that the approved test center can log in
#6	As a System Admin, I want to send billing to the test center so that they can make the payment
#7	As a System Admin, I want to view all clinic accounts so that I can see the clinic details
#8	As a System Admin, I want to search clinic account so that i can find a specific clinic
#9	As a System Admin, I want to filter by subscription plan to see which plan is the most popular among clinics.
#10	As a System Admin, I want to send recurring billing so that the clinics can make payment for the future months
#11	As a System Admin, I want to Suspend a clinic account so that they cannot log in
#12	As a System Admin, I want to unsuspend a clinic account so that they can log in
#13	As a System Admin, I want to view clinic detail so that I can know the clinic information
#14	As a System Admin, I want to view all product feedback so that I can improve on the product
#15	As a System Admin, I want to edit homepage so that I can make changes to the homepage
#16	As a System Admin, I want to view all FAQ so that I can see what are the questions posted there
#17	As a System Admin, I want to update FAQ so that the FAQ will be up to date and contains the latest information
#18	As a System Admin, I want to create new FAQ so that less people will ask the same question again
#19	As a System Admin, I want to view all user feedback so that I can see what needs to be changed or improved

#20	As a System Admin, I want to update billing plans so that it is on par to what we are offering
#21	As a System Admin, I want to update product features so that people can see the new changes or new information to the features
#22	As a System Admin, I want to update about us so that people can see the latest information of our product and teams
#23	As a System Admin, I want to update contact us so that people will have the latest details of our contacts
#24	As a System Admin, I want to log out of the system so that I can secure my account after use.

3.5.2 Use Case Description - System Admin

Name: Login (System Admin)		ID: A01
Stakeholders & Goals:	System Admin can successfully log into the system	
Description:	A System Admin needs to log in so that can use the system.	
Actors:	System Admin	
Pre-condition:	System admin had a valid account	
Trigger:	System admin clicks on the login button	
Normal Flow:	1. System admin types in username and password 2. System verifies with database 3. System displays the system admin homepage	
Sub-Flow:	None	
Alternative/Exceptional Flow:	1a. Incorrect username or password error is displayed	

Name: Logout (System Admin)		ID: A02
Stakeholders & Goals:	System Admin can successfully log out of the system	
Description:	The system admin logs out of the account	
Actors:	System Admin	
Pre-condition:	System Admin is already logged into the system.	
Trigger:	System Admin clicks the logout button	
Normal Flow:	1. System admin clicks the logout button 2. The system brings it back to the login page	
Sub-Flow:	None	
Alternative/Exceptional Flow:	None	

Name: View Pending Application (System Admin)		ID: A03
Stakeholders & Goals:	System admin to view all the submitted test centre applications	
Description:	To see the details of all the applied submission submitted by the test centre	
Actors:	System Admin	
Pre-condition:	System Admin is at the homepage	
Trigger:	System Admin clicks view pending application button	
Normal Flow:	1. System Admin clicks on the view submitted applications button 2. The system fetches all the submitted applications 3. The administrator clicks on one of the applications 4. The system displays the details of the application	
Sub-Flow:	None	
Alternative/Exceptional Flow:	1a. The system will show error message if no records are found	

Name: Approve Application (System Admin)		ID: A04
Stakeholders & Goals:	System admin to approve submitted applications	
Description:	An approve button to approve the application	
Actors:	System Admin	
Pre-condition:	System Admin is in each test centre application page	
Trigger:	System Admin clicks approve button	
Normal Flow:	1. System admin clicks on the approve button 2. The system marks and label it as approved 3. The system closes the application details page 4. The view test centre application page is displayed	
Sub-Flow:	None	
Alternative/Exceptional Flow:	None	

Name: Reject Application (System Admin)		ID: A05
Stakeholders & Goals:	System admin to reject submitted applications	
Description:	A reject button to reject the application	
Actors:	System Admin	
Pre-condition:	System Admin is in each test centre application page	
Trigger:	System Admin clicks reject button	
Normal Flow:	1. System admin clicks on the reject button 2. The system marks and label it as rejected 3. The system closes the application details page 4. The view test centre application page is displayed	
Sub-Flow:	None	
Alternative/Exceptional Flow:	None	

Name: Create Test Centre Admin Account (System Admin)		ID: A06
Stakeholders & Goals:	System admin to create account for test center admin	
Description:	Make an account for the approved test center admin for it to login and manage	
Actors:	System Admin	
Pre-condition:	System Admin is in the homepage	
Trigger:	System Admin clicks create account button	
Normal Flow:	1. System admin clicks create account button 2. System admin fills in the new account username and password 3. The system updates the new account to the database 4. "Account created successfully" notification is displayed 5. The system displays the system admin homepage	
Sub-Flow:	None	
Alternative/Exceptional Flow:	1a. Account creation failed is displayed if there is error writing to the database or invalid characters in the account username	

Name: Create new billing		ID: A07
Stakeholders & Goals:	System admin to send the bill to the test center	
Description:	A bill containing the amount with a due date emailed to the test center email address	
Actors:	System Admin	
Pre-condition:	System Admin is in each create test centre account page	
Trigger:	System Admin clicks view bill button	
Normal Flow:	1. System admin clicks view bill button 2. The bill is displayed 3. System admin clicks send bill button 4. System admin enters the email address of the test centre 5. The system sends the bill to the specified email 6. "Bill send successfully" is displayed	
Sub-Flow:	None	
Alternative/Exceptional Flow:	None	

Name: View All Clinic Account (System Admin)		ID: A08
Stakeholders & Goals:	System admin to view all the registered clinic account	
Description:	To see the click account and details	
Actors:	System Admin	
Pre-condition:	System Admin is in its homepage	
Trigger:	System Admin clicks View Clinics button	
Normal Flow:	1. The system admin clicks on the view clinics button 2. The system displays list of the clinics	
Sub-Flow:	None	
Alternative/Exceptional Flow:	1a. No clinics found error is displayed	

Name: Search Clinic Account (System Admin)		ID: A09
Stakeholders & Goals:	System admin to search a clinic	
Description:	To search a specific clinic by its name	
Actors:	System Admin	
Pre-condition:	System Admin is in view all clinics page	
Trigger:	System Admin clicks into the search bar	
Normal Flow:	1. The system admin clicks into the search bar 2. System admin types in the clinic name to search 3. The system does a database query with the specified name 4. The system displays the found clinic(s) in the search bar	
Sub-Flow:	None	
Alternative/Exceptional Flow:	1a. No result found is displayed when there are no matching clinic name in the database	

Name: Filter by Subscription Plan (System Admin) ID: A10	
Stakeholders & Goals:	System admin will select to group them by subscription tier
Description:	Clinics are classified and grouped together by their subscription plan
Actors:	System Admin
Pre-condition:	System Admin is in view all clinics page
Trigger:	System Admin clicks the dropdown menu beside the search bar
Normal Flow:	1. System admin clicks the dropdown menu and select the subscription plan tier 2. The system make a group query to the database 3. The system displays all the clinics in the selected subscription plan tier
Sub-Flow:	None
Alternative/Exceptional Flow:	1a. No clinics found in this subscription plan error is displayed when no clinics are associated with the specified plan

Name: Send Recurring Billing (System Admin)		ID: A11
Stakeholders & Goals:	System admin to send bill again for the following month	
Description:	To send the same bill to the client for the next month and so on	
Actors:	System Admin	
Pre-condition:	System Admin is in each clinic page	
Trigger:	System Admin clicks send recurring bill button	
Normal Flow:	1. System admin clicks send recurring bill button 2. The system automatically registered the clicked date, and schedule to send the bill every 30 days 3. "Recurring bill scheduled and sent successfully" message is displayed 4. The system displays the clinic page again	
Sub-Flow:	None	
Alternative/Exceptional Flow:	None	

Name: Suspend Clinic Account (System Admin)		ID: A12
Stakeholders & Goals:	System admin to suspend a clinic	
Description:	The suspended clinic cannot log in to the system	
Actors:	System Admin	
Pre-condition:	System Admin is in each clinic page	
Trigger:	System Admin clicks suspend button	
Normal Flow:	1. The system admin clicks on the suspend button 2. The system updates its database and flagged it as suspended 3. The clinic account page is displayed, with suspended clinic marked with a small red 'X' beside it	
Sub-Flow:	None	
Alternative/Exceptional Flow:	None	

Name: Unsuspend Clinic Account (System Admin)		ID: A13
Stakeholders & Goals:	System admin to unsuspend a clinic	
Description:	The suspended clinic can now log in to the system	
Actors:	System Admin	
Pre-condition:	System Admin is in each clinic page and the clinic must be suspended	
Trigger:	System Admin clicks unsuspend button	
Normal Flow:	1. The administrator clicks on the suspend button 2. The system updates its database and flagged it as unsuspended 3. The test centre account page is displayed, with the small red 'X' removed	
Sub-Flow:	None	
Alternative/Exceptional Flow:	None	

Name: View Clinic Details (System Admin)		ID: A14
Stakeholders & Goals:	System admin to see all the details of the clinic	
Description:	The details will includes the clinic email, location, phone number, description and subscription tier	
Actors:	System Admin	
Pre-condition:	System Admin is in each clinic page	
Trigger:	System Admin clicks on the clinic	
Normal Flow:	1. The system admin clicks on the clinic 2. The system displays the clinic details	
Sub-Flow:	None	
Alternative/Exceptional Flow:	None	

Name: View All Product Feedback		ID: A15
Stakeholders & Goals:	System admin to see feedback given by customers	
Description:	To see the feedback comments of our product and ways we can improve it	
Actors:	System Admin	
Pre-condition:	System Admin is in homepage	
Trigger:	System Admin clicks view feedback button	
Normal Flow:	1. The system admin clicks on the view feedback button 2. The system displays all the feedbacks	
Sub-Flow:	None	
Alternative/Exceptional Flow:	None	

Name: Edit Homepage (System Admin)		ID: A16
Stakeholders & Goals:	System admin to edit the system product homepage	
Description:	To customize the product homepage, make changes to the webpage layout or description	
Actors:	System Admin	
Pre-condition:	System Admin is in the homepage	
Trigger:	System Admin clicks on the edit layout button	
Normal Flow:	1. The system admin clicks the button to rearrange the layout 2. The system admin rearranges the layout 3. The system stores the layout in the database 4. The webpage reflects the updated layout	
Sub-Flow:	None	
Alternative/Exceptional Flow:	None	

Name: View All FAQ (System Admin)		ID: A17
Stakeholders & Goals:	System admin to see the list of faq	
Description:	System admin can see the faq for users if they have any questions on regards to navigating or using our products	
Actors:	System Admin	
Pre-condition:	System Admin is in the homepage	
Trigger:	System Admin clicks on the FAQ button	
Normal Flow:	1. The system admin clicks on the FAQ button 2. The system displays a list of FAQ items	
Sub-Flow:	None	
Alternative/Exceptional Flow:	None	

Name: Update FAQ (System Admin)		ID: A19
Stakeholders & Goals:	System admin to update individual FAQ	
Description:	To update the faq to the latest information	
Actors:	System Admin	
Pre-condition:	System Admin is in the FAQ page	
Trigger:	System Admin clicks on the edit button	
Normal Flow:	1. The system admin clicks on the edit button 2. The system admin types in the things to edit 3. The system admin clicks on the save button 4. The system updates the database 5. The system displays the list of updated faq page	
Sub-Flow:	None	
Alternative/Exceptional Flow:	1a. "Saving error" error message is shown if it contains invalid characters	

Name: Create New FAQ (System Admin)		ID: A20
Stakeholders & Goals:	System admin to create an FAQ	
Description:	To create an new FAQ for the question mostly asked by customers	
Actors:	System Admin	
Pre-condition:	System Admin is in the FAQ page	
Trigger:	System Admin clicks on the add new button	
Normal Flow:	1. The system admin clicks on the add new button 2. The system admin types in the things to add 3. The system admin clicks on the add button 4. The system updates the database 5. The system displays the list of added faq page	
Sub-Flow:	None	
Alternative/Exceptional Flow:	1a. Creating failed error message is shown if it contains invalid characters	

Name: View User Feedback (System Admin)		ID: A21
Stakeholders & Goals:	System admin to see all the feedbacks	
Description:	To see the feedbacks given by the users	
Actors:	System Admin	
Pre-condition:	System Admin is in the homepage	
Trigger:	System Admin clicks on the view feedbacks button	
Normal Flow:	1. The system admin clicks on the view feedbacks button 2. The System displays all the feedbacks	
Sub-Flow:	None	
Alternative/Exceptional Flow:	1a. No feedback found is shown when there is no feedbacks given to the specific user	

Name: Update Billing Plans (System Admin)		ID: A22
Stakeholders & Goals:	System admin to successfully retrieve current billing plans and make updates	
Description:	To changes the plans amount, add or remove features from different tier plans	
Actors:	System Admin	
Pre-condition:	System Admin is in the edit homepage page	
Trigger:	System Admin clicks on the edit button	
Normal Flow:	1. The system admin clicks on the edit button 2. The system admin edits the plans amount or description 3. The system admin clicks on the update button 4. The system updates its database with the plan 5. The system displays the new plans in the page	
Sub-Flow:	None	
Alternative/Exceptional Flow:	None	

Name: Update Product Features (System Admin)		ID: A23
Stakeholders & Goals:	System admin to successfully update the product features	
Description:	To change the product features, add new features	
Actors:	System Admin	
Pre-condition:	System Admin is in the edit homepage page	
Trigger:	System Admin clicks on the edit button	
Normal Flow:	The system admin clicks on the edit button The system admin edits the product features The system admin clicks on the update button The system updates its database with the features The system displays the new features in the page	
Sub-Flow:	None	
Alternative/Exceptional Flow:	None	

Name: Update About Us (System Admin)		ID: A24
Stakeholders & Goals:	System admin to successfully update the about us page	
Description:	To change the about us description	
Actors:	System Admin	
Pre-condition:	System Admin is in the edit homepage page	
Trigger:	System Admin clicks on the edit button	
Normal Flow:	The system admin clicks on the edit button The system admin edits the about us page The system admin clicks on the update button The system updates its database with the new information The system displays the new about us in the page	
Sub-Flow:	None	
Alternative/Exceptional Flow:	None	

Name: Update Contact Us (System Admin)		ID: A25
Stakeholders & Goals:	System admin to successfully update the contact us page	
Description:	To change the contact us details like phone number, email	
Actors:	System Admin	
Pre-condition:	System Admin is in the edit homepage page	
Trigger:	System Admin clicks on the edit button	
Normal Flow:	The system admin clicks on the edit button The system admin edits the contact us page The system admin clicks on the update button The system updates its database with the new information The system displays the new contact us in the page	
Sub-Flow:	None	
Alternative/Exceptional Flow:	None	

3.6 Use Case Diagram - Unregistered User



3.6.1 User Stories - Unregistered User

#1	As a Unregistered User, I want to register as a test center so that I can use this product in my test center
#2	As a Unregistered User, I want to submit application form so that i can be granted access to use this product
#3	As a Unregistered User, I want to register as a patient so that I can have a quicker diagnosis on my covid result
#4	As a Unregistered User, I want to add patient profile so that I can document it for future reference
#5	As a Unregistered Use , I want to view user feedback so that I can see what are the issues, bugs and improvements needed
#6	As a Unregistered User, I want to view billing plans so that I can see what features are offered at various plans
#7	As a Unregistered User, I want to view product statistics so that I have a clear understand of this product efficiency and performance
#8	As a Unregistered User, I want to view comparison in market so that I can see what features gives an competitive edge
#9	As a Unregistered User, I want to view product summary so that I can understand what is this product about and its usage
#10	As a Unregistered User, I want to view the FAQ so that it can answer to question that I have about this product
#11	As a Unregistered User, I want to a guided walkthrough so that I can understand how to navigate around the product and use each features
#12	As a Unregistered User, I want to have a contact us page so that potential clients can reach out to us
#13	As a Unregistered User, I want to have a about us page so that I can learn more about the people behind it, the team's vision and mission

3.6.2 Use Case Description - Unregistered User

Name: Register As A Test Center (Unregistered User)		ID: U01
Stakeholders & Goals:	Unregistered user to successfully register for an test center account	
Description:	To have an account with our product	
Actors:	Unregistered User	
Pre-condition:	Unregistered User is in the webpage homepage	
Trigger:	Unregistered User clicks on the register button	
Normal Flow:	1. The unregistered user fills in the details 2. The unregistered user clicks on the dropdown box of account type and select "test center" 3. The unregistered user clicks on the register button 4. The system adds new account to the database 5. The unregistered user is now logged in	
Sub-Flow:	None	
Alternative/Exceptional Flow:	1a. Password length error if password is less than 8 characters or it does not contains special characters	

Name: Register As A Patient (Unregistered User)		ID: U02
Stakeholders & Goals:	Unregistered user to successfully register for a patient account	
Description:	To have an account with our product	
Actors:	Unregistered User	
Pre-condition:	Unregistered User is in the webpage homepage	
Trigger:	Unregistered User clicks on the register button	
Normal Flow:	1. The unregistered user fills in the details 2. The unregistered user clicks on the dropdown box of account type and select "patient" 3. The unregistered user clicks on the register button 4. The system adds new account to the database 5. The unregistered user is now logged in	
Sub-Flow:	None	
Alternative/Exceptional Flow:	1a. Password length error if password is less than 8 characters or it does not contains special characters	

Name: Submit Application Form (Unregistered User)		ID: U03
Stakeholders & Goals:	Unregistered user to successfully submit application form	
Description:	To submit application form to access features of the product	
Actors:	Unregistered User	
Pre-condition:	Unregistered User is logged in to the account and the account type must be a test center account type	
Trigger:	Unregistered user clicks on submit application form	
Normal Flow:	1. Unregistered user clicks on submit application form button 2. Unregistered user fills up the form 3. The system stores the complete form in the database for review 4. A successfully submitted application message is shown 5. The system displays the unregistered user homepage	
Sub-Flow:	None	
Alternative/Exceptional Flow:	None	

Name: Add Patient Profile (Unregistered User)		ID: U04
Stakeholders & Goals:	Unregistered user to add a patient profile	
Description:	A profile consist of patient name, sex, age	
Actors:	Unregistered User	
Pre-condition:	Unregistered User is logged in to the account and the account type must be a patient account type	
Trigger:	Unregistered user clicks on add profile button	
Normal Flow:	1. Unregistered clicks on add profile button 2. The unregistered user fills up the profile section 3. Unregistered user clicks save 4. The system update the database with the new information 5. The profile updated message is shown	
Sub-Flow:	None	
Alternative/Exceptional Flow:	1a. Incorrect data type is entered, eg. age was given a character instead of a number 2a. The incompatible type error message is displayed	

Name: View User Feedback (Unregistered User)		ID: U05
Stakeholders & Goals:	Unregistered user to view other user's feedback on product	
Description:	To see what others had given their reviews on this product	
Actors:	Unregistered User	
Pre-condition:	Unregistered User is on the product homepage	
Trigger:	Unregistered user clicks on the view more button	
Normal Flow:	1. Unregistered user clicks on the view more button 2. The system displays all the user feedbacks	
Sub-Flow:	None	
Alternative/Exceptional Flow:	None	

Name: View Billing Plans (Unregistered User)		ID: U06
Stakeholders & Goals:	Unregistered user to view different billing plans	
Description:	To see what are the different plans and what features the product offers	
Actors:	Unregistered User	
Pre-condition:	Unregistered User is on the product homepage	
Trigger:	Unregistered user clicks on the view plans button	
Normal Flow:	1. Unregistered user clicks on the view plans button 2. The system displays all the plans available	
Sub-Flow:	None	
Alternative/Exceptional Flow:	None	

Name: View Product Statistics (Unregistered User)		ID: U07
Stakeholders & Goals:	Unregistered user to view product statistics	
Description:	To see our model prediction accuracy and performance	
Actors:	Unregistered User	
Pre-condition:	Unregistered User is at the product homepage	
Trigger:	Unregistered user clicks on the see features button	
Normal Flow:	1. Unregistered user clicks on the see features button 2. The system displays the details, description and charts	
Sub-Flow:	None	
Alternative/Exceptional Flow:	None	

Name: View Comparison In The Market (Unregistered User)		ID: U08
Stakeholders & Goals:	Unregistered user to view product comparison with other product	
Description:	To compare our features with other in the market	
Actors:	Unregistered User	
Pre-condition:	Unregistered User is at the product homepage	
Trigger:	Unregistered user clicks on the see features button	
Normal Flow:	1. Unregistered user clicks on the see features button 2. The system displays the details, description and charts 3. The unregistered user scroll down 4. The system then displays the features comparison table	
Sub-Flow:	None	
Alternative/Exceptional Flow:	None	

Name: View Product Summary (Unregistered User) ID: U09	
Stakeholders & Goals:	Unregistered user to view and understand our product summary
Description:	To have an overall understanding on our product and what we offer
Actors:	Unregistered User
Pre-condition:	Unregistered User is at the product homepage
Trigger:	Unregistered user clicks on the see features button
Normal Flow:	1. Unregistered user clicks on the see features button 2. The system displays the details, description and charts 3. The unregistered user scroll down 4. The system then displays the features comparison table 5. 5. The unregistered user scroll down 6. The system displays the product summary
Sub-Flow:	None
Alternative/Exceptional Flow:	None

Name: FAQ (Unregistered User)		ID: U10
Stakeholders & Goals:	Unregistered user to view product frequently asked question	
Description:	To clear any doubts or questions they might have with the product	
Actors:	Unregistered User	
Pre-condition:	Unregistered User is at the product homepage	
Trigger:	Unregistered user clicks on the help button	
Normal Flow:	1. Unregistered user clicks on the help button 2. The system displays all the FA questions at the top	
Sub-Flow:	None	
Alternative/Exceptional Flow:	None	

Name: Guided Tutorial / Walkthrough (Unregistered User)		ID: U11
Stakeholders & Goals:	Unregistered user to have a guided walkthrough on how to navigate around the product	
Description:	To show the user on how to do certain things, use certain features, which covers all the product features	
Actors:	Unregistered User	
Pre-condition:	Unregistered User is in the FAQ page	
Trigger:	Unregistered user clicks on the guided button	
Normal Flow:	1. Unregistered user clicks on the guided button 2. The system displays a step by step guide on how to navigate	
Sub-Flow:	None	
Alternative/Exceptional Flow:	None	

Name: Contact Us Page (Unregistered User)		ID: U12
Stakeholders & Goals:	Unregistered user to view the contact email, phone number, and a contact us form	
Description:	To view our contact details to be able to reach out to us or write us a question	
Actors:	Unregistered User	
Pre-condition:	Unregistered User is at the FAQ page	
Trigger:	Unregistered user click on contact us	
Normal Flow:	1. The unregistered user click on the contact us button 2. The system displays the contact us form 3. The system displays the contact details	
Sub-Flow:	None	
Alternative/Exceptional Flow:	None	

Name: About Us (Unregistered User)		ID: U13
Stakeholders & Goals:	Unregistered user to view the about us page	
Description:	To know about what we do, our vision and mission and the team behind it	
Actors:	Unregistered User	
Pre-condition:	Unregistered user is in the product homepage	
Trigger:	Unregistered User clicks on the about us button	
Normal Flow:	1. Unregistered user clicks on the about us button 2. The system displays the about us page	
Sub-Flow:	None	
Alternative/Exceptional Flow:	None	

4. Non-functional Requirements

4.1 Security Requirements

Security should ensure to provide confidentiality, integrity, and availability/authenticity. To provide confidentiality, all sensitive data such as passwords will be encrypted in the database. Integrity will be provided by using an integrity check and data validation functions to ensure that the data has not been modified. Login will be required prior to performing any actions by all users to provide availability/authenticity.

4.2 Reliability Requirements

Reliability shows how well the website is able to perform without having any failures and errors. With this, we must ensure that there will be minimal bugs in our system. Also, our website should be able to handle a certain amount of workload during usage.

4.3 Performance Requirements

Performance requirements define how well the software system accomplishes certain functions under specific conditions. In this case, the website should respond quickly as soon as the user navigates around the website. It should take no more than 5 seconds to load every page that the user navigates into. Additionally, the website should be able to be accessed by 400 users at the same time.

4.4 Maintainability Requirements

Maintainability defines the ease of fixing bugs and faults which are found within the software. Our code should allow new features and modifications to be implemented without hindering other functions of the software. This means the code should be neat whereby the admin/programmer can easily understand the structure and make respective changes.

4.5 Usability Requirements

Usability shows how easily the user is able to operate and interact with the site. Our team hopes to achieve high user satisfaction, for which we aim to make our site easily accessible to all, user-friendly with an appealing interface and easily navigable without the need to refer to a manual. Since we aspire to cover a large group of audience, we would try to make our website as secure and reliable as possible, in order to gain high customer loyalty.